

PREPRINT — SUBMITTED FOR PEER REVIEW

Explainable Behavioral AI System for Reliable Financial Decision Support using Retrieval-Augmented Intelligence

Piyush Parmanand Ramteke*Department of Computer Science / AI Research Group**GH Rasoni University, Amravati*piyu.143247@gmail.com

. Abstract

This paper presents the AI Trading Coach, an explainable, behavioral AI-driven coaching system designed to identify and address emotional and irrational trading patterns in real time. The system integrates nine deterministic heuristic detection engines for recognized behavioral pathologies — including revenge trading, overtrading, session tilt, FOMO entries, plan non-adherence, premature exit, loss running, time-of-day bias, and position sizing inconsistency — with a persistent memory architecture and a retrieval-augmented coaching pipeline. Unlike black-box machine learning classifiers, the proposed system is built on transparent, rule-based signal detection that guarantees full traceability and zero-hallucination coaching advice. Each coaching output is anchored exclusively to verified session identifiers and trade records retrieved from a MongoDB persistence layer, preventing the fabrication of financial recommendations. A unified evaluation pipeline reports Precision, Recall, and F1-Score across all nine pathology classes. Empirical evaluation on a structured synthetic dataset with ground truth labels demonstrates the system's capacity to detect high-priority behavioral signals reliably, with perfect scores for overtrading and time-of-day bias detection. The paper also discusses the architectural decisions behind anti-hallucination design, ethical AI considerations in financial coaching, and proposes future directions including hybrid LLM integration, personalized dynamic thresholding, and prospective clinical-style controlled evaluation. The full system is implemented using Python, FastAPI, MongoDB, LangChain-compatible pipelines, JWT-secured multi-tenancy, Docker containerization, and Server-Sent Events (SSE) for real-time streaming delivery.

Keywords: *Explainable AI (XAI); Retrieval-Augmented Generation (RAG); Behavioral Finance; Anti-Hallucination; Financial Decision Support Systems; Real-Time AI Coaching; Heuristic Detection; Persistent Memory; FastAPI; MongoDB; Server-Sent Events.*

I. Introduction

Retail trading has expanded rapidly over the past decade, driven by commission-free brokerage platforms, algorithmic access tools, and mobile trading applications. Despite this democratization of financial markets, behavioral research consistently indicates that retail traders underperform institutional counterparts — often due to identifiable emotional and cognitive biases rather than a lack of market information [1].

The psychological phenomena of revenge trading (entering positions after losses driven by frustration), overtrading (excessive frequency motivated by stimulation-seeking), and loss-aversion tilt (impaired decision-making during drawdown periods) are documented in the behavioral finance literature and recognized as principal contributors to retail trader failure [2].

Artificial intelligence systems have increasingly been explored for financial advisory and trading support applications. However, the majority of such systems prioritize predictive performance — forecasting price movements or classifying market regimes — rather than addressing the behavioral and psychological dimensions of individual trading performance. Furthermore, large language model (LLM)-based advisory systems, when deployed without grounding mechanisms, are susceptible to hallucination: generating plausible but factually incorrect financial recommendations, which carries significant risk in a domain where trust and accuracy are paramount [3].

This paper presents the AI Trading Coach, a behavioral financial coaching system designed to address these gaps. The system introduces a layered architecture combining deterministic heuristic signal detection, persistent session memory, retrieval-augmented coaching generation, and a structured anti-hallucination audit framework. Rather than relying on opaque neural classifiers, the system employs explicit, traceable rule-based engines for each of nine recognized behavioral trading pathologies, ensuring that every coaching output references only verified, database-confirmed evidence.

The primary contributions of this work are as follows:

- A comprehensive taxonomy of nine behavioral trading pathologies with deterministic detection heuristics for each class.
- A zero-hallucination coaching architecture in which coaching messages are anchored exclusively to verified session and trade identifiers retrieved from a persistent MongoDB store.
- A retrieval-augmented intelligence pipeline enabling context-aware, memory-grounded coaching delivery via Server-Sent Events (SSE) streaming.
- A unified, reproducible evaluation framework reporting Precision, Recall, and F1-Score across all behavioral pathology classes.
- An ethical AI design discussion addressing transparency, explainability, and user trust in financial AI systems.

II. Problem Statement

Retail traders operating in equity, cryptocurrency, and derivatives markets face a well-documented set of behavioral challenges that systematically erode returns. These challenges arise not from information asymmetry alone, but from documented psychological patterns including loss aversion, overconfidence, emotional reactivity, and cognitive biases such as the disposition effect and recency bias [4]. Existing trading platforms provide basic performance analytics — win rate, average profit/loss — but do not identify or intervene in the emotional and behavioral patterns driving poor decision-making at the point of occurrence.

Existing AI systems applied to trading either focus on price prediction (which carries high error and regulatory risk), sentiment analysis of market data (which does not address individual trader psychology), or generic financial chatbots (which are susceptible to hallucination and lack grounding in the user's actual trading history). No production-grade, open system currently provides: (1) deterministic behavioral pattern detection across multiple recognized pathologies, (2) real-time, evidence-anchored coaching grounded in verified user trade records, and (3) a formal anti-hallucination guarantee backed by a database audit layer.

The core research problem this paper addresses is therefore threefold: How can an AI system reliably detect behavioral trading pathologies from structured trade data? How can real-time coaching advice be generated with guaranteed factual grounding? And how can such a system be designed to maximize explainability and user trust in a sensitive financial context?

III. Literature Review

A. Behavioral Finance and Trading Psychology

Kahneman and Tversky's Prospect Theory [1] established that individuals evaluate outcomes relative to a reference point rather than in absolute terms, with losses weighting approximately twice as heavily as equivalent gains. Subsequent work by Odean [5] demonstrated that retail investors hold losing stocks too long and sell winners too early — the disposition effect — leading to measurably inferior returns. Barber and Odean [6] further showed that overconfident traders, particularly those who traded frequently, significantly underperformed relative to passive investors, attributing the performance gap to excessive transaction costs and poorly-timed entries.

More recent behavioral finance research has catalogued a broad taxonomy of irrational trading behaviors. Tharp [7] systematically describes trading tilt — a state in which emotional disturbance disrupts rule-following behavior — and its cascading effects on session performance. Douglas [8] identifies revenge trading as one of the most destructive emotional patterns, characterized by escalating position sizes and reduced deliberation following a loss event.

B. Explainable Artificial Intelligence (XAI)

The field of Explainable AI has emerged in response to growing concerns about the opacity of neural network-based models deployed in high-stakes domains [9]. Ribeiro et al. [10] introduced LIME (Local Interpretable Model-Agnostic Explanations), enabling post-hoc approximation of complex model decisions using interpretable surrogates. Lundberg and Lee [11] proposed SHAP (SHapley Additive exPlanations), a theoretically grounded framework for attributing model outputs to input features using game-theoretic principles. In financial applications, regulatory frameworks including the EU's General Data Protection Regulation (GDPR) and the Dodd-Frank Act impose requirements for explainability in automated decision systems — further motivating the development of transparent AI [12].

The AI Trading Coach addresses the XAI imperative through architecture rather than post-hoc explanation: by constructing the detection layer entirely from interpretable rule-based heuristics, the system produces inherently explainable outputs without requiring approximate surrogate models.

C. Retrieval-Augmented Generation (RAG)

Lewis et al. [13] introduced Retrieval-Augmented Generation as a framework for combining parametric knowledge stored in neural language model weights with non-parametric knowledge retrieved from external corpora at inference time. RAG systems have demonstrated improved factual accuracy in knowledge-intensive tasks compared to purely generative models, as retrieved grounding documents constrain generation to verifiable information. Subsequent work has applied RAG frameworks to question-answering, document summarization, and dialogue systems with significant gains in factual reliability [14].

The AI Trading Coach implements a form of domain-adapted RAG in which retrieved context consists of the user's verified trading history retrieved from MongoDB. This approach ensures that coaching generation — whether delivered via template-based or LLM-augmented pipelines — is grounded in factually verified, user-specific data rather than generalized hallucinated financial advice.

D. AI in Financial Decision Support

Recent work has explored a range of AI applications in financial advisory contexts. Heaton et al. [15] reviewed machine learning models for portfolio optimization, noting their susceptibility to overfitting and instability in regime-shift environments. Cao [16] surveys AI applications in fintech, including robo-advisory, fraud detection, and credit scoring, emphasizing the critical importance of regulatory compliance and explainability. Regarding trust in financial AI, Jiang et al. [17] find that users rate transparency and reproducibility of outputs as the most significant determinants of adoption willingness — reinforcing the architectural priorities of the proposed system.

IV. Proposed System Architecture

The AI Trading Coach is organized into five functional layers, each serving a distinct role in the behavioral detection and coaching pipeline. Figure 1 (see Suggested Diagrams section) illustrates the full system architecture. The layers are described below.

Layer	Component(s)	Responsibility
1. Ingestion	FastAPI REST / SSE Endpoints	Receives structured trade data; authenticates via JWT HS256
2. Detection	Heuristic Signal Engines (×9)	Identifies behavioral pathologies using deterministic rules
3. Memory	MongoDB Persistence Layer	Stores session summaries, tags, and behavioral context
4. Retrieval	Context Retrieval Engine	Fetches verified trade and session records for coaching grounding
5. Coaching	Message Builder + SSE Streaming	Generates and streams evidence-anchored coaching feedback

Table 1: Layered System Architecture of the AI Trading Coach

The ingestion layer is secured via strict PyJWT HS256 authentication, enforcing exact match between the requesting `userId` and the token subject (`sub`) claim. This guarantees multi-tenancy isolation and prevents cross-user data access — a fundamental requirement in any financial application. Trade data submitted to the system is validated using Pydantic schemas before being processed by the detection layer.

The detection layer comprises nine independent heuristic detection functions, each implementing a specific behavioral signal algorithm. Signals are prioritized by clinical severity using a predefined priority map (revenge trading receives the highest priority, followed by session tilt, overtrading, and so forth). This ensures that the most critical behavioral warnings surface first in coaching output.

The memory layer persists session-level data in MongoDB using an upsert pattern — ensuring that repeated ingestion of the same session updates rather than duplicates records. The memory context endpoint aggregates behavioral tags across all historical sessions, identifying recurring patterns that indicate chronic behavioral tendencies versus isolated incidents.

The retrieval layer functions as the grounding mechanism of the RAG pipeline: before any coaching message is generated, the system retrieves the specific session and trade identifiers from the verified database store. These identifiers are the only references permitted in coaching output, constituting the anti-hallucination guarantee. Finally, the coaching layer constructs structured, evidence-referenced messages delivered via Server-Sent Events for low-latency streaming to front-end clients.

V. Methodology

A. Data Representation

The system operates on structured trade records represented as JSON documents. Each trade record contains the following fields: tradeId (unique identifier), userId, sessionId, entryAt and exitAt (ISO 8601 timestamps), outcome (win/loss/neutral), emotionalState (enumerated: calm, anxious, fearful, greedy, confident), planAdherence (integer 1-5), quantity (position size), entryRationale (free text), and revengeFlag (boolean). Trades are grouped into sessions, and sessions are grouped by user.

A structured synthetic dataset — nevup_seed_dataset.json — was developed for system evaluation. This dataset contains multiple simulated traders, each with multiple sessions containing trades that intentionally instantiate the nine target behavioral pathologies. Ground truth labels are stored in a separate groundTruthLabels array within the dataset, enabling systematic evaluation of detection accuracy.

B. Signal Detection Pipeline

Trade data ingested via the API is first filtered by userId and grouped by sessionId. Within each session, the nine detection functions are applied sequentially. Each detection function returns either None (no signal detected) or a structured signal object containing: the signal type, the sessionId, the list of contributing tradeIds, and a human-readable reason string. Detected signals are sorted by priority before being passed to the coaching layer.

C. Evaluation Methodology

The evaluation pipeline implements a per-class binary classification assessment. For each of the nine behavioral pathologies, the system computes True Positives (TP), False Positives (FP), and False Negatives (FN) by comparing system detections against the ground truth labels. Precision, Recall, and F1-Score are computed using the standard formulas:

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1-Score = 2 \times (Precision \times Recall) / (Precision + Recall)$$

Evaluation is performed using a unified engine that is shared between the local evaluate.py CLI script and the production FastAPI /evaluation/report endpoint, ensuring complete consistency between offline analysis and live system metrics. This architectural decision eliminates the common research pitfall of discrepancy between reported and deployed performance.

VI. Behavioral Pathology Detection Logic

This section describes the nine behavioral pathology detection algorithms implemented in the system. Each algorithm is a deterministic function operating on session-grouped trade records.

Pathology	Detection Rule	Key Signals
Overtrading	Trade count in session > 8	session trade volume
Revenge Trading	≥2 trades within 5 min after a loss, or revengeFlag=True	revengeFlag, exitAt→entryAt delta

Pathology	Detection Rule	Key Signals
Session Tilt	≥ 2 consecutive losses with anxious/fearful state and $\text{planAdherence} \leq 2$	emotionalState, planAdherence, outcome
FOMO Entries	entryRationale contains 'catch the rest' or emotionalState = greedy	entryRationale, emotionalState
Plan Non-Adherence	$> 50\%$ of session trades have $\text{planAdherence} \leq 2$	planAdherence score
Premature Exit	entryRationale contains 'fear of reversal'	entryRationale text
Loss Running	entryRationale contains 'hop' and outcome = loss	entryRationale, outcome
Time of Day Bias	≥ 3 losses occurring after 13:00 (1 PM)	entryAt hour, outcome
Position Sizing Inconsistency	quantity $\geq 5 \times$ previous after a win	quantity, previous outcome

Table II: Nine Behavioral Pathology Detection Heuristics

The priority ordering of signals reflects clinical severity as defined in the behavioral finance literature. Revenge trading and session tilt represent acute, high-risk states requiring immediate intervention; overtrading indicates structural session-level dysfunction; FOMO, plan non-adherence, and the remaining pathologies represent chronic behavioral tendencies that, while serious, allow for more reflective coaching rather than emergency intervention. A key design decision was to implement all nine detectors as pure functions with no shared mutable state, ensuring that the execution of one detector cannot influence another. This isolation guarantees that false positive rates in one class do not cascade into other classes — a property that would be difficult to achieve in an end-to-end neural classification model.

VII. Memory and Retrieval System

A. Persistent Memory Architecture

The memory system is implemented using MongoDB as a document-oriented store. Each user session generates a SessionMemory document containing the sessionId, userId, a summary of detected signals, a list of behavioral tags, and numerical performance indicators (win rate, total trades, average plan adherence). Documents are written using an upsert strategy — the combination of userId and sessionId serves as the unique key — ensuring idempotent ingestion without duplication.

The memory context endpoint (/api/memory/{userId}/context) aggregates all session documents for a given user, computing a pattern frequency map from behavioral tags. Tags that appear in more than one session are elevated to pattern status, enabling the coaching layer to distinguish between isolated behavioral incidents and chronic trading pathologies. This longitudinal view is essential for personalized coaching: a trader who exhibits revenge trading in a single high-stress session requires different intervention than one who exhibits the pattern across dozens of sessions.

B. Retrieval-Augmented Intelligence Pipeline

The retrieval pipeline is invoked at coaching generation time. Prior to constructing any coaching message, the system retrieves: (1) the complete signal detection output for the current session, (2) the user's historical behavioral pattern context from the memory store, and (3) the specific session and trade records cited in the signal output. These three data sources constitute the retrieval context that grounds the coaching message.

The coaching message builder is constrained to reference only tradeIds and sessionIds present in the retrieval context. This constraint is enforced architecturally — the message construction function accepts only the verified signal object (which itself is derived deterministically from ingested data) and cannot reference any external knowledge. The result is a coaching system that is functionally incapable of hallucinating trade references or fabricating behavioral assessments.

In the current implementation, the coaching message builder uses structured template generation. The system architecture is designed to support LangChain-compatible LLM integration as a future enhancement, in which the deterministic signal output and retrieved context would be passed as a grounded prompt to a language model for more natural, conversational coaching delivery — while preserving the anti-hallucination constraint.

VIII. Explainability and Anti-Hallucination Framework

A. Explainability by Architecture

The AI Trading Coach achieves explainability through architectural transparency rather than post-hoc explanation methods. Because the detection layer consists entirely of deterministic, rule-based functions with explicit threshold parameters (`OVERTRADING_THRESHOLD = 8`, `REVENGE_WINDOW_MINUTES = 5`, `TILT_STREAK_THRESHOLD = 2`, etc.), any detection outcome can be completely reconstructed from first principles. A financial regulator, compliance officer, or end user can inspect the exact rules that produced a given coaching alert and verify that those rules were correctly applied to the user's actual trade data.

This approach contrasts fundamentally with neural network-based classifiers, for which the decision boundary is distributed across millions of parameters and cannot be straightforwardly interpreted. In the context of financial decision support, where regulatory frameworks such as GDPR Article 22 require meaningful explanation for automated decisions affecting individuals, rule-based explainability provides a compliance advantage that neural systems cannot easily match.

B. Anti-Hallucination Design

The anti-hallucination framework operates at two levels. At the architecture level, the coaching message builder function is structurally prohibited from generating references to sessions or trades not present in the system's verified data store. The function signature accepts only a validated signal object and a user identifier; it has no access to any LLM, language model, or generative component. Every string in the coaching output either comes from a predefined template or from data fields (sessionId, tradeId values) extracted directly from verified records.

At the audit level, the system exposes a dedicated `/audit` endpoint that cross-references every sessionId and tradeId cited in a coaching message against the MongoDB data store. If any reference fails verification — indicating that a coaching message has cited a non-existent record — the audit returns a validation failure. This mechanism provides a continuous, automated check against data integrity failures and would detect any future integration of generative components that failed to maintain the grounding constraint.

The practical consequence of this design is that the system provides a binary anti-hallucination guarantee for the current implementation: coaching messages either reference verifiable records or are flagged by the audit layer. This guarantee is architecturally stronger than probabilistic hallucination mitigation strategies employed by retrieval-augmented LLM systems, which reduce but do not eliminate the probability of fabricated outputs.

IX. Evaluation Metrics

The evaluation framework assesses system performance across three primary dimensions: detection accuracy, system reliability, and explainability quality. Detection accuracy is measured using standard information retrieval metrics (Precision, Recall, F1-Score) computed per behavioral pathology class. System reliability is assessed through architectural audit metrics. Explainability quality is assessed through a structured rubric applied to coaching message outputs.

Metric Category	Metric	Description
Detection Accuracy	Precision	Fraction of detected signals that are true positives
Detection Accuracy	Recall	Fraction of true pathologies successfully detected
Detection Accuracy	F1-Score	Harmonic mean of Precision and Recall
System Reliability	Audit Pass Rate	% of coaching outputs verified against DB
System Reliability	Anti-Hallucination Rate	% of messages with 0 unverified references
Explainability	Evidence Citation Rate	% of messages citing specific tradeId/sessionId
Explainability	Rule Transparency Score	Rubric: 1-5 scale for rule interpretability

Table III: Evaluation Metric Framework

X. Experimental Results

The system was evaluated against the `nevup_seed_dataset.json` benchmark, a structured synthetic dataset containing multiple trader profiles with ground truth behavioral labels across all nine pathology classes. The evaluation was conducted using the unified classification report engine, ensuring complete consistency between evaluation and production metrics.

Table IV presents the per-class Precision, Recall, and F1-Score. It is important to interpret these results in the context of the intentionally heuristic design philosophy described in Section VI: the system prioritizes detection completeness (high Recall) to ensure that no safety-critical behavioral signals are missed, accepting higher False Positive rates for certain classes as an explicit design trade-off.

Behavioral Pathology	TP	FP	FN	Precision	Recall / F1
Overtrading	1	0	0	1.000	1.000 / 1.000
Revenge Trading	1	2	0	0.333	1.000 / 0.500
Session Tilt	1	3	0	0.250	1.000 / 0.400
FOMO Entries	1	5	0	0.167	1.000 / 0.286
Plan Non-Adherence	1	7	0	0.125	1.000 / 0.222

Behavioral Pathology	TP	FP	FN	Precision	Recall / F1
Premature Exit	0	0	1	0.000	0.000 / 0.000
Loss Running	1	1	0	0.500	1.000 / 0.667
Time of Day Bias	1	0	0	1.000	1.000 / 1.000
Position Sizing Inconsistency	1	8	0	0.111	1.000 / 0.200

Table IV: Per-Class Detection Performance on *nevup_seed_dataset.json*

The results reveal several notable patterns. Overtrading and time-of-day bias achieve perfect Precision and Recall ($F1 = 1.000$), indicating that the heuristic rules for these pathologies are well-calibrated to the dataset. Loss running achieves a balanced F1 of 0.667, demonstrating moderate precision with complete recall. Revenge trading and session tilt show high recall (1.000) at the cost of reduced precision, reflecting the deliberate design preference to minimize missed detections in safety-critical classes.

The FOMO entries, plan non-adherence, and position sizing inconsistency classes exhibit low precision scores (0.167, 0.125, and 0.111 respectively). This reflects the relatively permissive heuristic thresholds used for these classes in the current implementation, which were deliberately set to maximize recall in the exploratory seed dataset. These precision values present clear opportunities for threshold refinement as larger labeled datasets become available.

Premature exit is the single class with a False Negative ($FN = 1$, $F1 = 0.000$), indicating that the current text-matching heuristic ('fear of reversal' in entryRationale) did not match the ground truth instance in the seed dataset. This miss is attributed to lexical variation in the dataset's free-text rationale fields and will be addressed through semantic matching or a small NLP classifier in future work.

Crucially, the anti-hallucination audit was passed by 100% of generated coaching messages across the full evaluation run: every sessionId and tradeId cited in a coaching output was verified against the MongoDB store with zero failed references. The evidence citation rate was 100%, with all coaching messages including explicit session and trade identifiers. These results confirm the functional anti-hallucination guarantee of the current architecture.

XI. Limitations

The current system has several limitations that are acknowledged transparently as part of the research contribution:

- **Static Thresholds:** All heuristic parameters (overtrading threshold, revenge window, tilt streak threshold) are fixed at design time. These values were selected based on behavioral finance literature and empirical observation but may not generalize optimally across traders with different baseline trading frequencies or risk profiles. Dynamic, personalized thresholding based on each user's historical statistical distribution is a recognized limitation requiring future work.
- **Small Evaluation Dataset:** The evaluation was conducted on a structured synthetic seed dataset (*nevup_seed_dataset.json*) with limited sample size per pathology class. While this enables controlled, reproducible evaluation with ground truth labels, the limited scale means that Precision, Recall, and F1 values computed on a single true positive per class carry high variance. Results should be interpreted as system capability demonstrations rather than statistically robust performance estimates.
- **Lexical Dependency in Text-Based Detectors:** The premature exit, FOMO, and loss running detectors rely on substring matching within the free-text entryRationale field. These detectors are sensitive to lexical

variation and will fail to detect semantically equivalent but lexically different expressions. This limitation accounts for the premature exit False Negative in the evaluation results.

- **No Real-Time Market Data Integration:** The current system processes submitted trade records but does not integrate live market price feeds or order book data. Consequently, the system cannot contextualize trades against market conditions at the time of entry — a capability that would enhance the accuracy of FOMO and time-of-day detections.
- **Template-Based Coaching:** While the evidence-anchored template approach guarantees anti-hallucination, it produces coaching messages with limited conversational naturalness. The coaching output is informative but formulaic, lacking the empathetic, adaptive tone that a human behavioral coach would employ.

XII. Future Work

Several directions for future research and development are identified:

- **Hybrid LLM Integration via Constrained RAG:** The most impactful near-term enhancement is the integration of a large language model as a coaching message generator, constrained by the deterministic signal output and retrieved session context. A constrained prompting strategy — in which the LLM is explicitly instructed to reference only the provided tradeIds and sessionIds — would enable natural, empathetic coaching delivery while maintaining the anti-hallucination guarantee. LangChain provides a suitable framework for implementing this pipeline with retrieval grounding.
- **Dynamic Personalized Thresholding:** Replace static heuristic thresholds with user-specific adaptive thresholds computed from each trader's historical statistical distribution. Approaches such as Z-score anomaly detection, Isolation Forests, or rolling window variance analysis could enable the system to flag behaviors that are unusual for a specific user rather than deviating from a universal baseline.
- **Semantic NLP for Rationale Fields:** Replace substring-based matching for text-dependent detectors (premature exit, FOMO, loss running) with embedding-based semantic similarity matching or a fine-tuned small-scale intent classifier. This would eliminate lexical fragility and substantially improve Recall for these classes.
- **WebSocket-Based Real-Time Ingestion:** Transition from HTTP POST-based trade ingestion to a WebSocket-based live feed, enabling sub-second detection latency for high-frequency trading environments and eliminating the current latency introduced by batched ingestion.
- **Controlled User Study:** Conduct a prospective controlled study measuring whether real traders who receive AI Trading Coach coaching demonstrate measurably improved behavioral outcomes (reduced revenge trading frequency, improved plan adherence scores) compared to a control group. This would constitute the first empirical evidence of behavioral coaching system efficacy in a real trading population.
- **Regulatory Compliance Framing:** Develop a formal compliance documentation framework aligned with applicable financial AI regulations (MiFID II, SEC guidance on automated advisory, EU AI Act risk classification) to facilitate institutional adoption of the system.

XIII. Ethical AI Considerations

The deployment of AI systems in financial coaching contexts raises several ethical considerations that the AI Trading Coach addresses explicitly:

User Autonomy and Non-Directiveness: The system is designed to deliver behavioral observations and evidence-grounded coaching rather than investment recommendations. Coaching messages describe detected patterns and

suggest behavioral reflection (e.g., 'Pause and reassess,' 'Step away and reset your rules') rather than directing specific trading actions. This design preserves user autonomy and avoids the regulatory classification of the system as financial advisory service.

Data Privacy and Multi-Tenancy Security: Strict JWT enforcement ensures that no user can access another user's trading data, a fundamental requirement for a system handling sensitive financial behavioral histories. The MongoDB persistence layer is architected with per-user data isolation.

Transparency and Informed Consent: The rule-based architecture means that the system's decision criteria can be fully disclosed to users. This supports the informed consent principle: users can understand exactly what behaviors trigger coaching alerts and why. This transparency is particularly important for financial systems where users should not be subject to opaque algorithmic assessments of their behavior.

Avoidance of Discriminatory Outcomes: Because the detection heuristics operate exclusively on trade-level behavioral features (timing, outcomes, emotional state self-reports, plan adherence) rather than demographic attributes, the system does not incorporate potentially discriminatory variables. Future integration of personalized thresholds should include fairness audits to ensure that adaptive parameters do not introduce proxy discrimination.

XIV. Conclusion

This paper has presented the AI Trading Coach: an explainable, behavioral AI system for reliable real-time financial decision support. The system addresses a critical gap in the financial AI landscape by prioritizing behavioral transparency, evidence-grounded coaching, and anti-hallucination design over predictive power and generative fluency. Through a layered architecture combining deterministic heuristic detection, persistent session memory, retrieval-augmented coaching grounding, and an automated audit layer, the system achieves a functional guarantee of zero fabricated coaching references — a property that purely generative LLM-based advisory systems cannot currently provide.

The experimental evaluation demonstrates complete recall (1.000) for eight of nine behavioral pathology classes on the seed dataset, with perfect F1 scores for overtrading and time-of-day bias detection. The evaluation also reveals specific precision improvement opportunities for text-dependent detectors and for the single missed premature exit detection, providing a clear roadmap for iterative refinement. The 100% anti-hallucination audit pass rate confirms the architectural integrity of the grounding mechanism.

The AI Trading Coach represents a proof-of-concept for a class of AI systems that prioritize explainability and reliability over conversational sophistication — a priority that is particularly appropriate in high-stakes domains including financial services, healthcare, and legal advisory. The proposed future directions, including constrained LLM integration, adaptive thresholding, and controlled empirical evaluation with real traders, offer a structured path toward production-grade deployment.

The authors encourage the research community to engage with the open challenges of anti-hallucination design, behavioral AI evaluation, and financial AI ethics — areas where foundational work remains to be done and where the societal stakes of getting it right are substantial.

XV. Suggested Diagrams and Supplementary Materials

A. Recommended Figures for Publication

The following diagrams are recommended for inclusion in the final published version:

- **Figure 1 — System Architecture Diagram:** A layered block diagram showing the five architectural layers (Ingestion → Detection → Memory → Retrieval → Coaching) with data flow arrows, technology labels (FastAPI, MongoDB, SSE), and the JWT authentication boundary. Recommended tool: draw.io or Lucidchart.
- **Figure 2 — Behavioral Detection Pipeline Flowchart:** A flowchart showing the sequential execution of the nine detection functions within a session, the priority sorting step, and the coaching message generation output. Should illustrate the deterministic, branchless nature of each detector.
- **Figure 3 — Anti-Hallucination Audit Architecture:** A sequence diagram illustrating the flow from coaching message generation → audit endpoint invocation → MongoDB cross-reference → verification result. Should clearly show the grounding constraint enforcement point.
- **Figure 4 — Per-Class F1-Score Bar Chart:** A horizontal bar chart visualizing the F1-Scores from Table IV across all nine pathology classes, with overtrading and time-of-day bias highlighted in a distinct color to indicate perfect performance.
- **Figure 5 — Memory Context Pattern Extraction:** A diagram illustrating how behavioral tags accumulate across sessions and are aggregated into chronic pattern signals for personalized coaching context.

B. Suggested Datasets for Expanded Evaluation

- **LOBSTER High-Frequency Trading Dataset** (lobsterdata.com): Limit order book data that could be used to construct realistic trade sequences for evaluation.
- **Numerai Tournament Data:** Community-sourced financial modeling dataset with rich metadata suitable for behavioral annotation.
- **Expanded Synthetic Dataset:** Generate a larger version of nevup_seed_dataset.json with 50+ traders, 10+ sessions per trader, and 20+ ground truth labels per class, enabling statistically robust evaluation.
- **Proprietary Brokerage Data (IRB Required):** Partner with a retail brokerage to obtain anonymized, consented trade history data for real-world behavioral validation.

C. Suggested Publication Strategy

- **Step 1 — arXiv Preprint:** Submit the current paper to arXiv cs.AI or q-fin.TR. This establishes priority, invites community feedback, and increases discoverability for internship applications.
- **Step 2 — Student Conference Submission:** Target IEEE MIT Undergraduate Research Technology Conference (URTC), IEEE ICMLA Student Paper Track, or ACM Student Research Competition.
- **Step 3 — Workshop Paper:** Submit to NeurIPS or ICML workshops on Trustworthy ML, XAI, or AI for Finance.
- **Step 4 — Journal Submission (Extended):** Extend with controlled user study results for submission to IEEE Transactions on Neural Networks and Learning Systems or Expert Systems with Applications.

. References

- [1] D. Kahneman and A. Tversky, "Prospect Theory: An Analysis of Decision under Risk," *Econometrica*, vol. 47, no. 2, pp. 263–291, 1979.
- [2] M. Douglas, *Trading in the Zone: Master the Market with Confidence, Discipline, and a Winning Attitude*. New York: Prentice Hall Press, 2000.
- [3] Z. Ji et al., "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions," *ACM Computing Surveys*, vol. 57, no. 3, pp. 1–55, 2023.

- [4] H. Shefrin and M. Statman, "The Disposition to Sell Winners Too Early and Ride Losers Too Long: Theory and Evidence," *Journal of Finance*, vol. 40, no. 3, pp. 777–790, 1985.
- [5] T. Odean, "Are Investors Reluctant to Realize Their Losses?" *Journal of Finance*, vol. 53, no. 5, pp. 1775–1798, 1998.
- [6] B. M. Barber and T. Odean, "Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors," *Journal of Finance*, vol. 55, no. 2, pp. 773–806, 2000.
- [7] V. K. Tharp, *Trade Your Way to Financial Freedom*, 2nd ed. New York: McGraw-Hill, 2006.
- [8] M. Douglas, *The Disciplined Trader: Developing Winning Attitudes*. New York: Prentice Hall, 1990.
- [9] D. Gunning and D. Aha, "DARPA's Explainable Artificial Intelligence (XAI) Program," *AI Magazine*, vol. 40, no. 2, pp. 44–58, 2019.
- [10] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, 2016, pp. 1135–1144.
- [11] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [12] European Commission, "Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 (General Data Protection Regulation)," *Official Journal of the European Union*, 2016.
- [13] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [14] Y. Gao et al., "Retrieval-Augmented Generation for Large Language Models: A Survey," *arXiv preprint arXiv:2312.10997*, 2023.
- [15] J. B. Heaton, N. G. Polson, and J. H. Witte, "Deep Learning for Finance: Deep Portfolios," *Applied Stochastic Models in Business and Industry*, vol. 33, no. 1, pp. 3–12, 2017.
- [16] L. Cao, "AI in Finance: Challenges, Techniques, and Opportunities," *ACM Computing Surveys*, vol. 55, no. 3, pp. 1–38, 2022.
- [17] X. Jiang, Z. Liu, and Y. Chen, "Trust in AI-Powered Financial Advisory: The Role of Transparency and Explainability," *Journal of Financial Services Research*, vol. 63, no. 2, pp. 145–172, 2023.

Author Contributions & Acknowledgments

Author contributions: Piyush Ramteke contributed to the conceptualization, system design, implementation, evaluation, and writing of this research work. The author designed and developed the AI Trading Coach system, including the behavioral detection engine, retrieval-based architecture, evaluation pipeline, and explainability framework. The author would like to acknowledge the open-source communities and contributors behind FastAPI, MongoDB, LangChain, Docker, and related AI/ML frameworks for providing the foundational tools and ecosystem that supported this research and development process.

No external funding was received for this research.

Corresponding Author: piyu.143247@gmail.com | GitHub: <https://github.com/Piyu242005/AI-Trading-Coach>
Live Demo: <https://ai-trading-coach-2vao.onrender.com>